World's smallest computer is here!

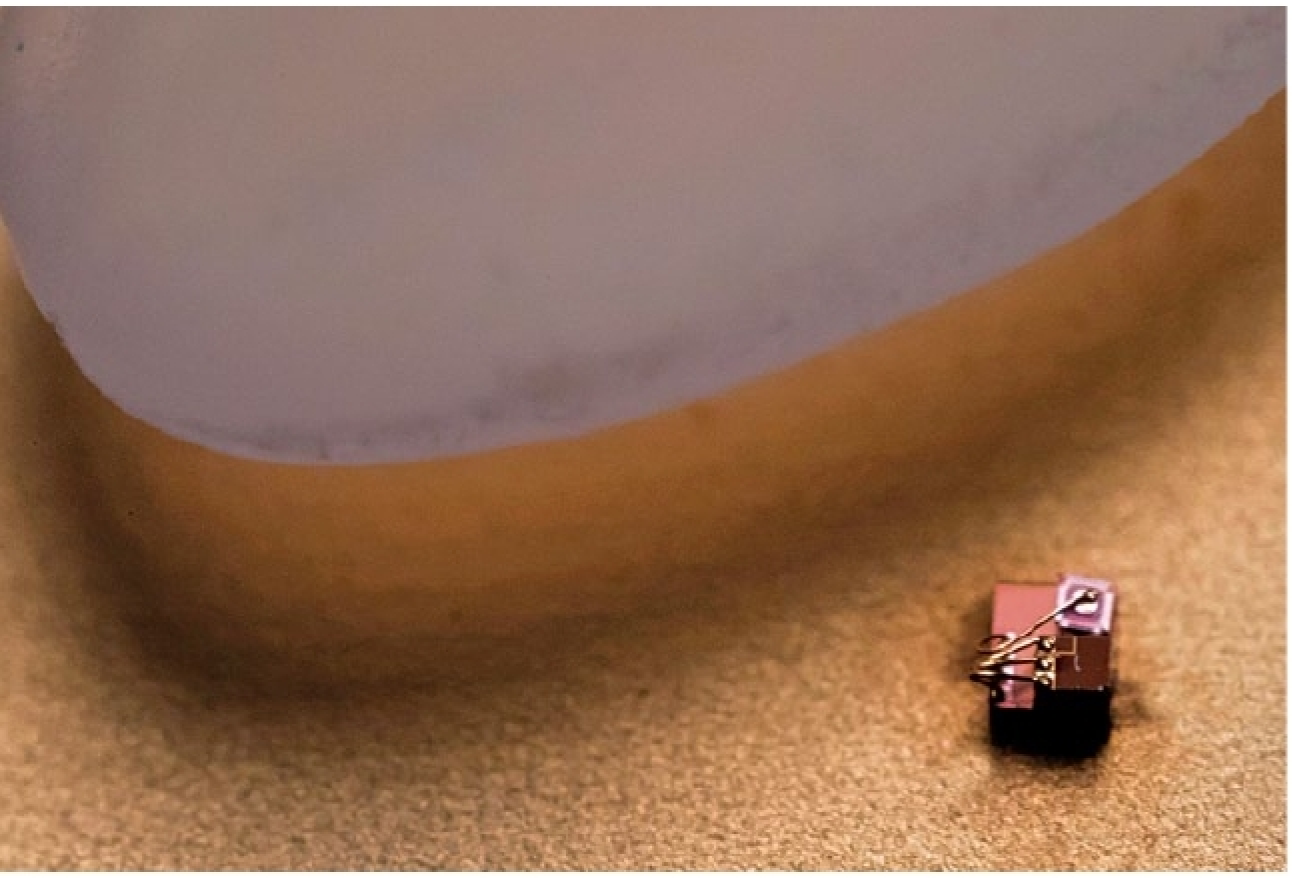
When IBM announced in March that it had produced the world's smallest computer, it raised a few eyebrows at University of Michigan, which is home to the previous champion of tiny computing. Now, the team at Michigan University has developed an even smaller device, measuring just 0.3mm to a side—smaller than a grain of rice.

IBM has called for a re-examination of what constitutes a computer. Previous systems, including 2x2x4mm Michigan Micro Mote, retain their programming and data even when these are not externally powered. Unplug a desktop computer, and its program and data are still there when it boots itself up once the power is back. These new microdevices, from IBM and now Michigan, lose all prior programming and data as soon as they lose power.

In addition to RAM and photovoltaics, the new computing devices have processors and wireless transmitters and receivers. Because these are too small to have conventional radio antennae, these receive and transmit data with visible light. A base station provides light for power and programming, and it receives data.

One of the big challenges in making a computer about one-tenth the size of IBM's was figuring out how to run at very low power when the system packaging had to be transparent. Light from the base station—and from the device's own transmission LED—can induce currents in its tiny circuits.

"We basically had to invent new ways of approaching circuit design that would be equally low power but could also tolerate light," said David Blaauw, professor of electrical and computer engineering, who led the development of the new system together with Dennis Sylvester, also professor of ECE, and Jamie Phillips, an Arthur F. Thurnau Professor and professor of ECE. That meant exchanging diodes, which can act like tiny solar cells, for switched capacitors.



Michigan Micro Mote (M3) next to a grain of rice for scale (Credit: Michigan Engineering)

Another challenge was achieving high accuracy while running on low power, which makes many of the usual electrical signals (like charge, current and voltage) noisier. Designed as a precision temperature sensor, the new device converts temperatures into time intervals, defined with electronic pulses. Intervals are measured on-chip against a steady time interval sent by the base station and then converted into a temperature. As a result, the computer can report temperatures in minuscule regions, such as a cluster of cells, with an error of about 0.1°C.

The system is flexible and could be reimagined for a variety of purposes, but the team chose precision temperature measurements because of a need in oncology. Their long-standing collaborator, Gary Luker, professor of radiology and biomedical engineering, wants to answer questions about temperature in tumours.

Robot that can sort recycling by giving it a squeeze

Scientists at Computer Science and Artificial Intelligence Lab at Massachusetts Institute of Technology (MIT) have developed a robot arm with soft grippers that picks up objects from a conveyor belt and identifies what these are made from by touch.

The robot, called RoCycle, uses capacitive sensors in its two pincers to sense the size and stiffness of the materials it handles. This allows it to distinguish between different metal, plastic and paper objects. In a mock recycling-plant setup, with